

## CSCE 638 NATURAL LANGUAGE PROCESSING · MODULE 12 · LECTURE NOTES

# Bias, Failure, and Fabrication

*How these systems fail people.*

## MODULE OUTCOMES

- Explain where bias enters and why it is not one thing.
- Explain distributional and dialectal failure.
- Explain fabrication and its mechanism.
- Explain what mitigations do and do not achieve.
- Build a failure analysis into a system's evaluation.

## 1 Where bias enters

| Stage      | What enters there                            | Example                                                                                       |
|------------|----------------------------------------------|-----------------------------------------------------------------------------------------------|
| Corpus     | Who writes, and about whom.                  | Web text skews heavily by language, region, era, and voice — and the skew is not random.      |
| Tokeniser  | Which languages are represented efficiently. | Several times more tokens for the same content in some scripts (Module 02 §3).                |
| Objective  | What gets treated as normal.                 | The majority pattern becomes the prediction, and the minority pattern becomes an error.       |
| Annotation | Whose judgement becomes the label.           | Raters disagree systematically by group, and the majority label erases the disagreement.      |
| Evaluation | Which failures are visible at all.           | Aggregate metrics hide subgroup failure — see the note.                                       |
| Deployment | Who bears the cost of an error.              | The same error rate with unequal consequences, depending on who can appeal or work around it. |

The evaluation row is the one that makes all the others invisible. An aggregate score can improve while a particular subgroup's experience gets worse — and it will not show up anywhere — **which is why disaggregation is the minimum requirement** (§2's callout) and why this row is the one to fix first: it is cheap, and it makes the rest measurable.

**“BIAS” NAMES SEVERAL DIFFERENT THINGS, AND CONFLATING THEM PREVENTS FIXING ANY OF THEM**

**A representational association** — the embedding geometry linking occupations to genders (Module 04 §3) — **is a property of the representation**, measurable with a similarity computation.

**A performance disparity** — higher error rates for one dialect, language, or name distribution — **is a property of the model's accuracy, and is measurable per group** given a labelled sample (§2).

**An allocation harm** — a deployed system's decisions distributing opportunity, attention, or risk unequally — **is a property of the deployment rather than of the model at all**, and cannot be measured by examining the model.

**And they require different responses. Measuring the first tells you very little about the third** — which is **why a debiased embedding can sit quite happily inside a system that allocates unfairly**, and why debiasing-the-vectors research answered a narrower question than it was taken to answer (Module 04 §3's post-hoc correction warning).

## 2 Distributional failure

- **Dialect. Error rates are measurably higher for non-standard varieties** — because the training data contains less of them, **and because the annotation frequently treats dialectal forms as errors**, which encodes the judgement into the label.
- **Language. Quality falls sharply outside the best-resourced languages, and the tokenisation penalty compounds the data scarcity** (Module 02 §3) — so the same question costs more and is answered worse.
- **Names and entities. Recognition is measurably worse for names underrepresented in the corpus** — which **affects exactly the people least able to work around it**, since the workaround is usually to change how you write your own name.
- **Domain and register. Clinical notes, legal drafting, and informal conversational text all differ substantially from the pretraining mixture**, and the degradation is not always obvious because the output remains fluent.
- **And the common cause across all four is the training distribution** — which means **the fix is data, and data is expensive. That is the honest statement**, and it is **why these disparities persist across model generations and are not solved by scale alone**: scaling a skewed corpus preserves the skew.

**WHICH MAKES DISAGGREGATED EVALUATION THE MINIMUM REQUIREMENT**

**Report performance by subgroup rather than only in aggregate** — because **an aggregate number averages over precisely the disparity you need to see**, and an improvement in the mean is consistent with a regression for a minority of users.

**And the subgroups have to be chosen from who actually uses the system**, which requires knowing that — **and is a question to ask at design time rather than after deployment**, since the data to answer it has to be collected.

**With the sample size per group large enough to measure anything**, which **frequently means deliberately collecting more data for small groups rather than sampling proportionally** — a proportional sample gives you a wide confidence interval exactly where you most need a narrow one.

**Which is CSCE 633 Module 12's subgroup argument** — and **it is the single most actionable thing in this module**, because **it converts a concern into a number that somebody can be held accountable for**, which is the step at which this material stops being a discussion and becomes engineering.

### 3 Fabrication

**THE MODEL WAS TRAINED TO PRODUCE PLAUSIBLE TEXT, AND FALSE TEXT CAN BE ENTIRELY PLAUSIBLE**

Next-token prediction optimises for what text looks like (Module 03 §1's last point) — and a well-formed false statement looks exactly like a well-formed true one, token by token, which means the objective cannot distinguish them even in principle.

There is no internal mechanism separating them. The model has no representation of 'I do not know this' that reliably governs its output, because nothing in the training process required one — and the probing evidence for partial internal signals of uncertainty does not yet translate into reliable behaviour.

And preference training makes it worse in a specific and predictable way (Module 08 §4): **confident, fluent, well-formatted answers rate better than hedged ones**, so the training explicitly rewards the behaviour that causes the harm. This is a Goodhart failure (CSCE 701 Module 12) inside the training pipeline.

Which is why fabrication is structural rather than a bug to be fixed — and why it is most dangerous precisely where the model is most fluent, which is also, unfortunately, where people trust it most. The correlation runs the wrong way.

- **Retrieval with citations** (Module 11) — the most effective available mitigation, because it makes the claim *checkable* rather than making it true. That distinction is the whole value, and it is why Module 11 §1 insists the citations are not optional.
- **Abstention training and explicit instruction**, which reduces the rate and does not eliminate it — and must be measured rather than assumed (Module 11 §4's third evaluation).
- **Calibration and confidence reporting**, which helps a reader weight the answer appropriately — and these models are poorly calibrated on open-ended output, so the reported confidence is itself of limited reliability.
- **Verification against a source**, which genuinely works and costs a second system plus the latency — and is the right answer where the cost of a wrong answer is high.
- **And designing so that a wrong answer is recoverable** — which is the only mitigation that does not depend on the model's behaviour at all. This is the architectural answer: put the output where a person checks it before it matters, which is CSCE 701 Module 11's consequence-reduction argument applied here, and the same move as designing so that being phished does not matter.

### 4 Building it into the evaluation

- **Performance disaggregated by the subgroups who will actually use it, with the sample sizes stated** so a reader can see which comparisons are meaningful (§2).
- **The input types on which it fails, found by error analysis** (Module 10 §4) rather than predicted — because the predicted failures and the actual ones overlap less than anyone expects.

- **The fabrication rate on questions your sources cannot answer**, measured deliberately by constructing such questions — which is ten minutes of work and is almost never done.
- **The confident-error rate, separately from the overall error rate** — because **a wrong answer the system is sure about is the one that causes harm**, and it is invisible in an aggregate accuracy figure.
- **And what happens downstream when it is wrong — who notices, how quickly, and what it costs. This last item is the one that turns this from an ethics section into an engineering requirement, and it is a design question rather than a measurement:** the answer is determined by where you put the output, not by how good the model is.

### THE HONEST FRAMING

**These systems are genuinely useful and they fail unevenly** — and **both halves have to be stated together**, because either one on its own is misleading: the first alone oversells, and the second alone implies they should not be used.

**The failures are not random: they concentrate on the data that the training distribution underrepresented**, which means **they concentrate on particular people** rather than being spread evenly across users.

**And the mitigations are partial. Nothing in §3 eliminates fabrication, and nothing in §2 equalises performance** — so **the deployment has to be designed on the assumption that both continue**, rather than on the assumption that the next model fixes them.

**Which makes the central design question 'what happens when this is wrong, and to whom'** — and **a system deployed without an answer to that question has not been evaluated**, whatever its benchmark scores. **That is Module 13's position**, and it is this course's version of the program's closing rule.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                    | What it covers for this module                                                                                       |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Bender et al. — On the Dangers of Stochastic Parrots (free) | <b>§1 and §2's argument</b> , and the paper that set the terms of this discussion. Read it in full.                  |
| Blodgett et al. — Language (Technology) is Power (free)     | <b>§1's callout, argued carefully</b> — a survey of how 'bias' is used inconsistently and what that costs.           |
| Ji et al. — Survey of Hallucination in NLG (free)           | <b>§3's mechanisms and mitigations</b> , organised — and honest about how partial the mitigations are.               |
| Mitchell et al. — Model Cards for Model Reporting (free)    | <b>§4's documentation practice</b> — the template that made disaggregated reporting a norm rather than an exception. |

## Exercises

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Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **For your Project 2 system, name what enters at each of the six stages.**
- 2 **Distinguish the three senses of bias** for one concrete concern you have.
- 3 **Measure performance on standard and non-standard text** for one task.
- 4 **Measure token counts** for the same content in three languages.
- 5 **Disaggregate one evaluation by a subgroup** and report the sample sizes.
- 6 **Construct ten questions your sources cannot answer** and measure the fabrication rate.
- 7 **Measure the confident-error rate** separately from the overall one.
- 8 **Add citations to one system** and check whether they support the claims.
- 9 **Design one recovery path** so that a wrong answer is caught before it matters.
- 10 **Write the two-sentence honest framing** for your own system.

## Self-check

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Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the six stages and what bias enters at each.
- 2 Why is the evaluation stage the one to fix first?
- 3 Distinguish the three senses of 'bias' and say why it matters.
- 4 Give four measured disparities and their common cause.
- 5 Why does scale not solve them?
- 6 What is the minimum requirement, and why is it the most actionable?
- 7 Give the mechanism of fabrication, and why preference training worsens it.
- 8 Why is it most dangerous where the model is most fluent?
- 9 Name five mitigations and say what the most effective one actually does.
- 10 Give the five components of a failure analysis.